

GIANLUCA GUGLIELMO

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RESEARCH FOCUS

Trustworthy computer vision, specifically focusing on the challenges and limitations of Out-of-Distribution detection.

EDUCATION

PhD Student, Insitute of Visual Computing

2023 - Current

Technical University Graz

Research topic: Out-of-Distribution Detection; joint supervision of Dr. Marc Masana and Prof. Thomas Pock.

M.Sc. in Data Science & Scientific Computing

2020 - 2023

University of Trieste

Final Grade: 110/110

B.Sc. in Physics

2016 - 2020

University of Trieste

Final Grade: 94/110

EXPERIENCE

R&D Engineer

Infineon

Oct. 2022 - Nov. 2023

Villach, Austria

Master Thesis Student

Infineon

Feb. 2022 - Sep. 2022

Villach, Austria

Teaching Assistant

University of Trieste

Nov. 2021 - Jan. 2022

Trieste, Italy

Computer Vision Intern

Video Systems Srl

Jan. 2019 - Apr. 2019

Codroipo, Italy

PUBLICATIONS

Ranked Activation Shift for Post-Hoc Out-of-Distribution Detection

G. Guglielmo, M. Masana

European Conference of Computer Vision, 2026

We studied the limitations of SoTA auxiliary OoD methods and introduced RAS, a consistent, model-agnostic, auxiliary method that matches the average feature profile of ID data.

Leveraging Intermediate Representations for Better Out-of-Distribution Detection

G. Guglielmo, M. Masana

Proceedings of the 28th Computer Vision Winter Workshop (CVWW 2025), 2025

We generalized the definition of energy to intermediate layers and showed that they can, on average, discriminate between ID and OoD better than the penultimate or last layers.

Automated Classification of Semiconductor Defect Density SEM Images Using Deep Learning

C. Kofler, F. L. Rosa, D. Zarre, **G. Guglielmo**, C. A. Dohr, J. Dohr, A. Zernig, A. Fernández-Caballero

Recent Advances in Microelectronics Reliability, Springer, Cham, pp. 99–116, 2024

Work done while at Infineon, it is a study on the detection of microscopic defects found on silicone wafers, captured with scanning electron microscopes.

WORKSHOPS

5th CLVISION Workshop @ CVPR2024 - Seattle

Challenge organizer at the 5th CLVISION Workshop.